

Sushma Ravirala

Data Analyst

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SUMMARY

Senior Data Analyst with 5+ years of experience leveraging SQL, Python, Power BI, and advanced analytics to optimize healthcare operations and drive actionable insights. Expert in healthcare claims analytics, patient utilization analysis, fraud detection, and readmission management, with a proven track record of improving operational efficiency, data accuracy, and patient care outcomes. Skilled in designing end-to-end ETL pipelines, building interactive dashboards, implementing predictive models, and collaborating with clinical and care management teams to enhance decision-making. Adept at standardizing multi-source healthcare datasets, aligning KPIs with business goals, and ensuring regulatory compliance, consistently delivering measurable improvements in healthcare delivery and reporting accuracy.

SKILLS

- Data Analysis & Programming:** SQL (SQL Server, PostgreSQL, Oracle), Python (Pandas, NumPy, Scikit-learn, Matplotlib, Seaborn), R (statistical modeling, visualization), SAS, Excel (Pivot Tables, Power Query, Advanced Formulas)
- Data Visualization & Reporting:** Power BI, Tableau, Excel Dashboards, KPI Design, Interactive Reports, DAX, Data Storytelling
- Data Engineering & ETL:** Data Warehousing (Snowflake, Azure SQL DW), ETL Development, Data Pipelines (Airflow, SSIS), Data Quality Frameworks, Data Validation & Cleansing, Data Standardization
- Statistical & Predictive Analytics:** Regression Analysis, Clustering, Time Series Forecasting, Predictive Modeling, Outlier Detection, Hypothesis Testing, Risk Stratification, A/B Testing
- Healthcare Domain Expertise:** Healthcare Claims Analytics, Patient Utilization Analysis, Fraud Detection in Claims, Readmission Analysis, Cost Optimization, Healthcare KPIs (LOS, Readmission Rate, Bed Utilization), Medical Coding Standards (ICD-10, CPT, DRG)
- Business & Functional Skills:** Requirements Gathering, Stakeholder Collaboration (Clinical, Care Management, Operations), Business Process Improvement, KPI Alignment, Regulatory Compliance (HIPAA, Data Governance), Root Cause Analysis
- Tools & Platforms:** Azure, AWS, Git, JIRA, Confluence, Agile/Scrum, Version Control, Automation Workflows

WORK EXPERIENCE

Humana, USA | Senior Data Analyst Jun 2023 – Present

- Engineered an end-to-end healthcare claims data quality and reporting framework, leveraging SQL, Python, and Power BI to validate and transform complex claims datasets while ensuring HIPAA compliance, improving claim error detection speed and enhancing operational efficiency for claims processing teams.
- Developed predictive models and clustering analyses to identify high-risk members and prevent repeated ER visits, collaborating with care coordination teams to reduce unnecessary healthcare utilization by 15% through actionable insights.
- Built interactive dashboards and automated reporting systems to track claim trends, member utilization, and prescription adherence, providing leadership with real-time visibility and improving operational efficiency while aligning KPIs with healthcare operations.
- Collaborated with clinical and care management teams to analyze patient utilization and optimize care pathways, identifying opportunities to shift appropriate cases from inpatient to outpatient settings and enhance preventive care programs, resulting in a 20% improvement in efficient healthcare delivery.
- Implemented robust data pipelines and automated validation rules for multi-source healthcare datasets, ensuring consistent data integrity, regulatory compliance, and streamlined decision-making for business teams and operations.

Persistent Systems, India | Data Analyst Feb 2020- Mar 2022

- Orchestrated the integration of patient claims, admissions, and discharge data into a unified warehouse using SQL and Python, which streamlined healthcare reporting pipelines and enhanced accuracy across cross-functional teams.
- Engineered automated data cleaning and validation frameworks to remove duplicates, standardize medical codes, and flag anomalies, which elevated claims accuracy and improved patient care monitoring effectiveness.
- Designed and deployed interactive Power BI dashboards that visualized fraud trends, readmission rates, and occupancy patterns while ensuring HIPAA compliance, empowering healthcare administrators to make faster, evidence-based decisions.
- Analyzed patient claims and readmission data using exploratory and statistical techniques, identifying operational bottlenecks and providing actionable insights that reduced fraudulent claims by over 18% and improved readmission outcomes by over 12%.
- Delivered actionable insights by transforming raw data into predictive and descriptive analytics solutions, which enabled hospitals to improve operational efficiency by over 25% and increase patient satisfaction by over 20%.

Mphasis, India | Junior Data Analyst Jun 2019 – Jan 2020

- Supported the creation of a healthcare data validation framework by cleansing and standardizing multi-source patient and claims datasets, which enhanced data accuracy and improved error detection by over 35% through systematic preparation and analysis.
- Developed automated workflows for medical claims and patient record checks, ensuring HIPAA compliance, streamlining reporting processes, and enabling clinical teams to quickly identify discrepancies, reducing manual intervention.
- Built interactive Power BI dashboards to monitor patient admission trends, claim approval accuracy, and anomaly detection, empowering healthcare administrators to make evidence-based decisions and strengthen compliance.
- Collaborated with senior analysts to document validation rules for ICD-10 and CPT codes while maintaining data lineage, which ensured reproducibility of results and increased overall team efficiency by over 25%.
- Monitored and reconciled high-volume patient admission and claims datasets, detecting fraudulent claim patterns and inconsistencies, providing actionable insights that improved transparency and operational outcomes.

EDUCATION

Bachelor of technology in computer science engineering May 2019

JNTUH Hyderabad, India